

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE
NAME

CLASS

CHEMISTRY

Paper 2

8872/02

Monday 26 August 2013

2 hours

Candidates answer Section A on the Question Paper

Additional Materials: Answer Paper

Data Booklet

Graph Paper (2 sheets)

READ THESE INSTRUCTIONS FIRST

Write your name and HT group on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** the questions

Section B

Answer **two** questions on separate answer paper.

You are advised to spend not more than 1 hour for Section B.

You are advised to show all working in calculations.

You may use a calculator.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Paper 1	30
Section A	40
A1	/10
A2	/5
A3	/15
A4	/10
Section B	40
B5	/20
B6	/20
B7	/20
TOTAL	110

This document consists of **14** printed pages and **0** blank page.

[Turn over

Section A

Answer **all** questions in this section in the spaces provided.

- 1** Al_2O_3 is significant in its use to produce aluminium metal and as a refractory material due to its high melting point.

a) i) Aluminium oxide is also said to be *amphoteric*. Explain the term in italics.

.....

ii) Write equations to show that aluminium oxide is amphoteric.

.....

.....[3]

- b)** When dry chlorine is passed over a piece of heated aluminium foil in a hard glass tube, a vapour (M_r : 267) is produced which condenses to a yellow-white solid in the cooler parts of the tube.

i) Suggest the molecular formula of the vapour and draw the structure of the molecule, showing clearly the bonds involved.

The yellow-white solid reacts with water in two different ways.

Reaction 1: When a few drops of water are added to the solid, steamy fumes are evolved and a white solid remains, which is insoluble in water.

Reaction 2: When a large amount of water is added to the solid, an acidic solution is obtained.

ii) Suggest the identities of the steamy fumes and white solid remaining at the end of **reaction 1**.

.....

iii) Write equations, including state symbols, for **reaction 2** and state the type of reaction that has occurred.

.....

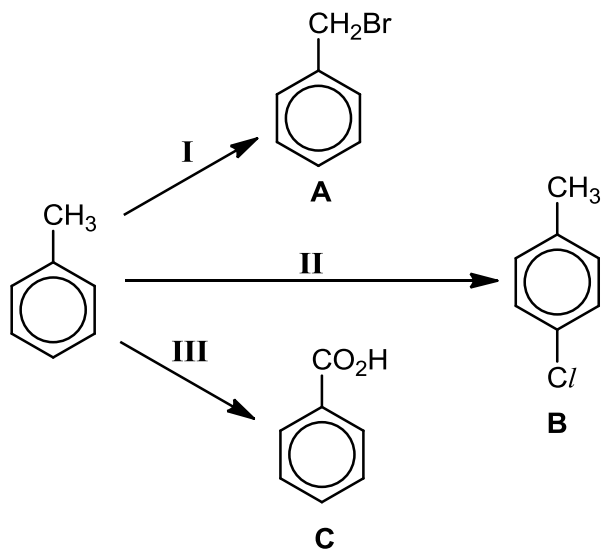
.....

Type of reaction:..... [7]

[Total: 10]

2 Methylbenzene is a useful starting material for the synthesis of many intermediates used to make pharmaceuticals, dyes and agrochemicals.

- a) Suggest the reagents and conditions needed for each of the reaction I, II and III in the following scheme.



Reaction I :

Reaction II :

Reaction III :

[3]

- b) Suggest a simple chemical test to distinguish between compounds **A** and **C**. In your answer, state clearly the reagents and condition used and the observations you will make.

.....

[2]

[Total: 5]

- 3 The standard enthalpy change of atomisation of an element or compound, ΔH_{atom} , is defined as the enthalpy change **when 1 mole of separate gaseous atoms** is formed from its elements or compounds under standard conditions.

For example, the standard enthalpy change of atomisation of carbon is represented as:

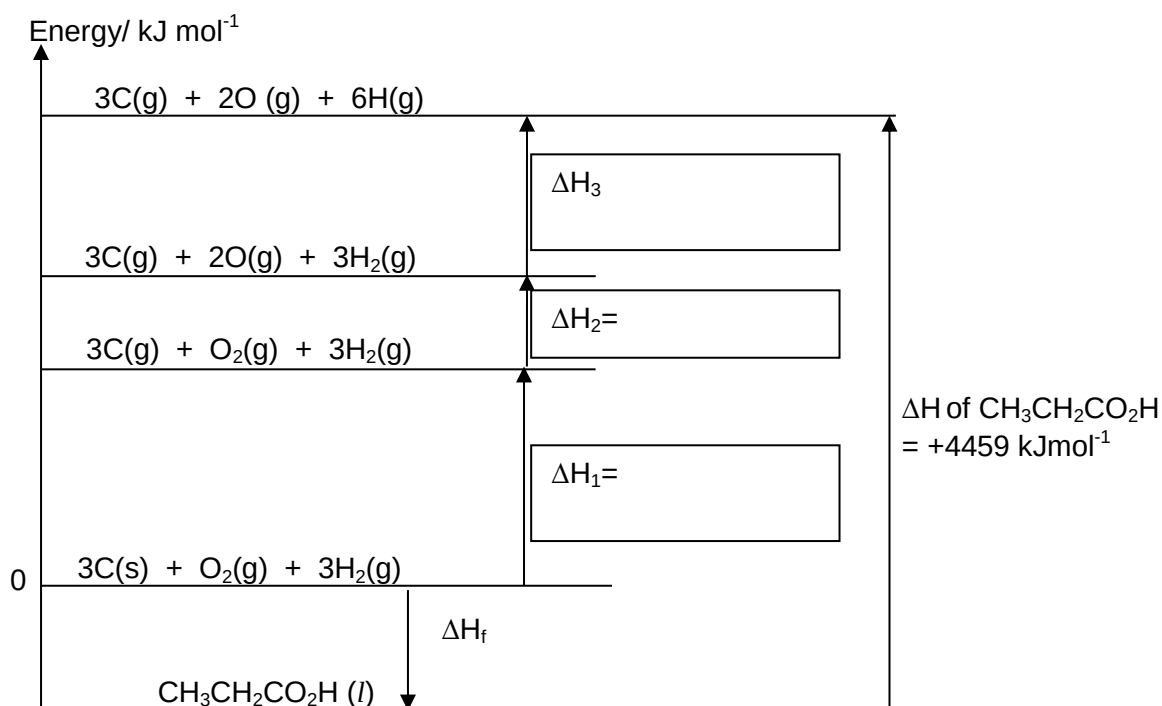


Bond energies are often used to calculate the enthalpy change of atomisation of simple covalent molecules.

- a) i) Define bond energy of dissociation.

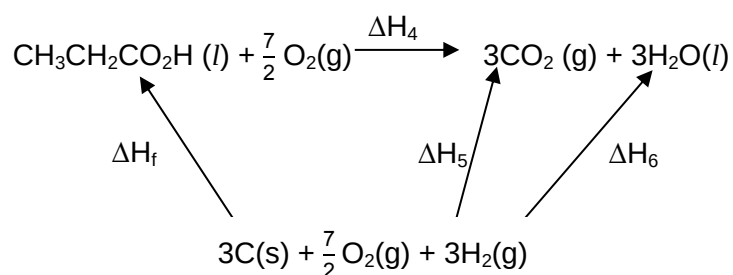
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- ii) Using data from the *Data Booklet* and value of ΔH_{atom} given above, complete the energy level diagram below by calculating ΔH_1 , ΔH_2 and ΔH_3 . Hence calculate the standard enthalpy change of formation of propanoic acid, $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$, showing your workings clearly.



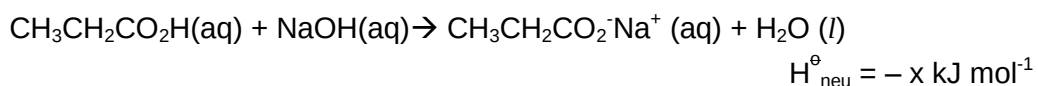
- b) Alternatively, the standard enthalpy change of formation of propanoic acid can be calculated using an energy cycle. Using the data given below, find ΔH_4 , ΔH_5 and ΔH_6 . Hence calculate the standard enthalpy change of formation of propanoic acid. Show clearly your working.

Compounds	Standard enthalpy changes / kJ mol^{-1}
$\text{CH}_3\text{CH}_2\text{CO}_2\text{H}(l)$	$\Delta H_c = -1527.2$
$\text{H}_2\text{O}(l)$	$\Delta H_f = -285.8$
$\text{CO}_2(g)$	$\Delta H_f = -393.5$



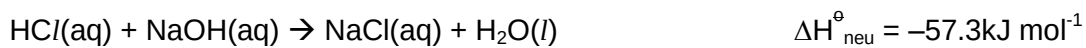
[2]

- c) Propanoic acid reacts with sodium hydroxide according to the following equation:



Where $\Delta H_{\text{neu}}^\ominus$ is the enthalpy change of neutralisation for the reaction.

Hydrochloric acid reacts with sodium hydroxide in a similar way:

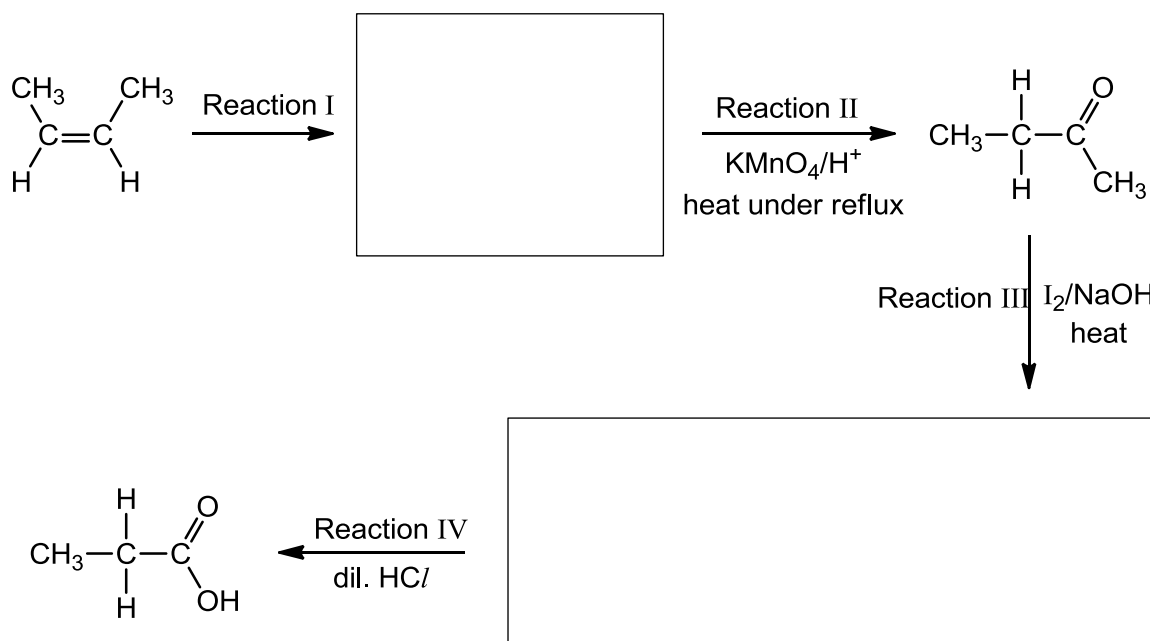


State how would the $\Delta H_{\text{neu}}^\ominus$ of propanoic acid with sodium hydroxide compared to that of hydrochloric acid with sodium hydroxide. Explain your answer.

.....

 [2]

- d) Propanoic acid can be synthesised in the laboratory from but-2-ene. The reaction scheme is shown below:



- i) Draw in all the organic compounds formed from reaction I and III.

- ii) State the reagent and condition in reaction I.

.....

- iii) State the type of reaction for the following:

Reaction I:

Reaction III:

- iv) Write a balanced equation for reaction III.

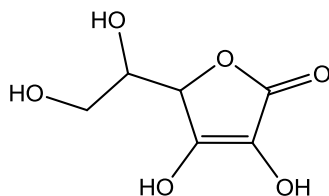
.....

[6]

[Total: 15]

- 4 Ascorbic acid, commonly known as Vitamin C, is an important water-soluble vitamin and biological antioxidant. Vitamin C is needed by the body for the growth and repair of bones, muscles, blood vessels as well as aiding in the absorption of iron.

Vitamin C has the following structure:



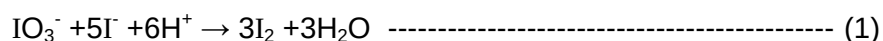
To determine the concentration of ascorbic acid in a Vitamin C tablet, an experiment known as *iodometric titration* is used. Firstly, Vitamin C is reacted with a known amount of iodine that is present in excess. Vitamin C will reduce the brown iodine present to colourless iodide. Secondly, unreacted excess iodine is then titrated against a standard solution of sodium thiosulfate. Once the amount of sodium thiosulfate is used is known, the amount of vitamin C present originally can then be determined.

To prepare a solution of ascorbic acid for the experiment, one 500 mg Vitamin C tablet is dissolved in sulfuric acid and diluted with deionised water up to 250 cm³ mark in a standard flask. 25 cm³ of this ascorbic acid solution is pipetted into a conical flask containing 3.00 × 10⁻⁴ mol of iodine solution (excess). Lastly, the resulting solution in the conical flask is immediately titrated with a standard sodium thiosulfate solution until the brown colour of iodine has lightened considerably to a pale yellow.

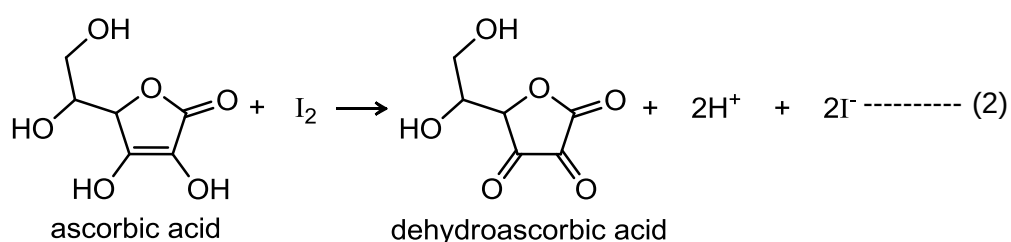
Starch indicator was added at this point. Starch forms an intense dark blue colour with the iodine still present in the conical flask. The titration with sodium thiosulfate is continued until the dark blue colour disappears leaving a colourless solution. The sudden disappearance of the dark blue colour is the distinct colour change that marks the end-point of the titration.

Reaction 1:

Solutions of iodine at fixed concentrations are difficult to prepare and store. Hence iodine will be generated *in situ* by the reaction of known amount of potassium iodate with an excess of potassium iodide in the presence of sulfuric acid as shown below:

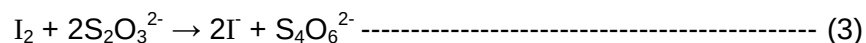


The iodine formed will oxidise Vitamin C, C₆H₈O₆, to form dehydroascorbic acid



Reaction 2:

The excess iodine will react with a standardised sodium thiosulfate solution.



It was found 12 cm³ of 0.005 mol dm⁻³ sodium thiosulfate is required to observe the disappearance of the dark blue colour.

- a)** Explain why Vitamin C is considered as a water-soluble vitamin.

.....
[1]

- b)** With reference to the species involved, describe the type of reaction occurring in equation 1.

.....

[2]

- c) i)** Calculate the amount of iodine that reacted with sodium thiosulfate.

- ii)** Calculate the amount of iodine that reacted with 25 cm³ of ascorbic acid solution.

iii) Calculate the amount of ascorbic acid in the 250 cm³ standard solution.

iv) Determine the mass percentage of ascorbic acid in a 500 mg Vitamin C tablet (M_r of ascorbic acid is 176).

[4]

d) Suggest a reason why the resulting solution is immediately titrated.

.....[1]

e) Suggest the reagent and conditions that could be used to distinguish between ascorbic acid and dehydroascorbic acid. State the observations.

.....
.....
.....[2]

[Total:10]

Section B

Answer **two** questions from this section on separate answer sheet.

- 5 An organic compound **A** has the following composition by mass: C, 68.3%; H, 6.4%; Cl, 25.3%. It has a relative molecular mass of 140.5.

When compound **A** is heated under reflux with aqueous KOH, compound **B** is formed.

Heating compound **B** with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ followed by immediate distillation gives compound **C**.

When 2,4-dinitrophenylhydrazine reagent is added to compound **C**, orange crystals were produced. Fehling's solution gives a brick red precipitate when warmed with compound **C**.

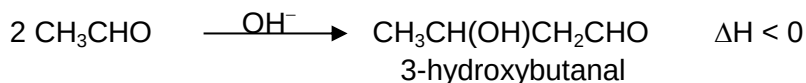
Heating compound **B** under reflux with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ gives compound **D**.

When compound **D** is reacted with solid Na_2CO_3 , a colourless gas which produces white precipitate in aqueous calcium hydroxide is produced.

- a) Calculate the molecular formula of compound **A**. [2]
- b) Use all the above information, including your answer in **a)**, to identify the organic compounds **A**, **B**, **C** and **D**, and draw a structural formula for each compound. State the reactions and write equations for all the reactions described above. Clearly show **all** of the deductions that you make from the information that you have been given; full marks cannot be awarded by only giving the structures required. [14]
- c) An isomer of compound **C** gives silver mirror when heated with Tollens' reagent. However, it does not give brick red precipitate when heated with Fehling's solution. Suggest a possible structure of this isomer. [1]
- d) When heating compound **B** under reflux with acidified $\text{K}_2\text{Cr}_2\text{O}_7$, a small amount of organic compound **E** (M_r 240), is produced.
 - i) Suggest a possible structure of **E**.
 - ii) State the reaction that has occurred. [2]
- e) Compound **A** can also be produced when compound **B** is reacted with anhydrous PCl_5 . With the aid of an equation, explain why PCl_5 used must be anhydrous. [1]

[Total: 20]

- 6 a) Under the influence of a dilute base or acid, two molecules of an aldehyde or a ketone may combine through the formation of carbon-carbon bonds in a process known as aldol reaction. An example of an aldol reaction between ethanal molecules in dilute alkaline solution is shown below.



A chemist studied the kinetics of the above reaction at 298 K and obtained the following table of results.

Experiment Number	Initial $[\text{CH}_3\text{CHO}]$ / mol dm^{-3}	Initial $[\text{OH}^-]$ / mol dm^{-3}	Initial rate / $\text{mol dm}^{-3} \text{ min}^{-1}$
1	0.10	0.010	0.013
2	0.30	0.010	0.039
3	0.20	0.030	0.078

- i) Using the above data, deduce the order of reaction with respect to each of the reagents, showing how you arrive at your answers. Hence, write the rate equation for the above reaction.
- ii) Calculate the rate constant, stating its units.
- iii) The effect of temperature on the rate constant, k , can be explained using the Arrhenius Equation.

$$k = Ae^{-E_a/RT}$$

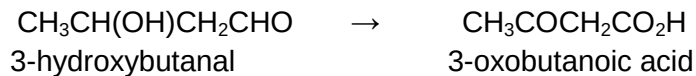
where E_a is the activation energy and both A and R are constants and T is the temperature.

Using the above equation, state the effect on k when the temperature is raised to 348 K.

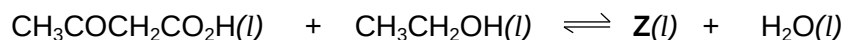
- iv) Sketch a fully labeled reaction pathway diagram illustrating the effect of addition of catalyst in the above aldol reaction.
- v) With reference to the Maxwell Boltzmann Theory, explain why the addition of catalyst speeds up a chemical reaction. [10]

b) 3-hydroxybutanal can be converted to 3-oxobutanoic acid via the following reaction.

i) Suggest the type of reaction, reagents and conditions required for this conversion.



ii) In the presence of concentrated sulfuric acid, 3-oxobutanoic acid reacted with ethanol to form an ester, **Z**.



The numerical value of the equilibrium constant is 4. When 1.7 mol of $\text{CH}_3\text{COCH}_2\text{CO}_2\text{H}$ and x mol of $\text{CH}_3\text{CH}_2\text{OH}$ is placed in a 2 dm^3 vessel at room temperature, it is found that the equilibrium mixture contains 1.6 mol of the ester.

- I Draw the skeletal formula of **Z** formed.
- II Find the no. of mol of ethanol, x .
- III With reference to Le Chatelier's Principle, explain why ethanol is used in excess.
- IV Explain the role of sulfuric acid in the formation of **Z**. [10]

[Total: 20]

- 7 In 1669, the German alchemist, Hennig Brand, discovered phosphorus, which can exist as white phosphorus and red phosphorus in its elemental form.

- a) Silicon, phosphorus and sulfur are consecutive elements in period 3 of the periodic table. Explain why the first ionisation energy of phosphorus is higher than that of both silicon and sulfur. [4]

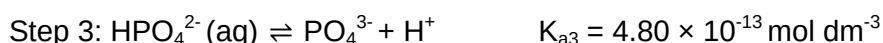
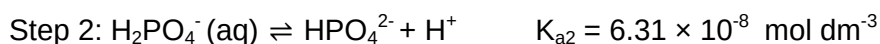
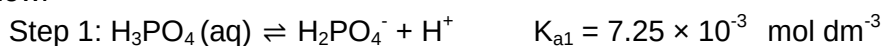
Phosphorus compounds are commonly used in commercial products. One example is calcium phosphide, Ca_3P_2 , which is used in fireworks and rat poisons.

- b) Describe with the aid of a labeled diagram, how the paths of separate beams of ions in Ca_3P_2 are affected on passing through an electric field at right angles of their direction of travel. Indicate clearly the relative magnitude and direction of deflection. [2]

Another commonly used commercial product is phosphorus oxoacids such as phosphorus acid, H_3PO_3 , and phosphoric acid, H_3PO_4 .

- c) i) State the oxidation state of phosphorus in H_3PO_3 .
 ii) Given that the shapes of the H_3PO_3 and tetrachloromethane molecules are the same, draw the dot-and-cross diagram for a molecule of H_3PO_3 . Hence, state the bond angle around phosphorus.
 iii) Would you expect H_3PO_3 to be more soluble in water or in an inert organic solvent such as tetrachloromethane? Explain your answer. [6]

- d) Phosphoric acid is a weak tribasic acid. It dissociates into PO_4^{3-} in the three steps shown below.



- i) Write an expression for K_{a1} .
 ii) Using the information given above, identify the strongest conjugate base. Explain your answer. [2]
 e) A student performed a titration by adding 10 cm^3 of NaOH with a concentration of $0.150 \text{ mol dm}^{-3}$ to 25 cm^3 of phosphoric acid with a concentration of $0.200 \text{ mol dm}^{-3}$ and claimed that the resultant solution is an acidic buffer.
 i) Define an acidic buffer.
 ii) Suggest a suitable indicator for the above titration. State the colour change expected at the end point of the titration.
 iii) Do you agree with the student's claim? Explain your answer with relevant calculations. You may assume that the H^+ is only produced in the first two dissociation steps of phosphoric acid. [6]

[Total: 20]